

PRECOG Lab - NLP Recruitment Task 2026

IIIT Hyderabad

The Ghost in the Machine

AI vs Human Text Classification

"Le style, c'est l'homme même"

"The style is the man himself." - Georges-Louis Leclerc

Submitted by

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Page 1: Task Completion Summary

As required by PRECOG Lab, this page details exactly what parts of the task were completed and what parts were not, along with reasons.

Tasks Completed

Task	Description	What I Did	Status
Task 0	Dataset Creation	Downloaded Dickens + Austen novels, extracted 15 topics, generated 1500 paragraphs (3 classes)	COMPLETE
Task 1	Statistical Fingerprinting	Extracted 13 linguistic features (TTR, hapax, POS, dependency depth, readability, punctuation)	COMPLETE
Task 2 - A	XGBoost Classifier	Trained on 13 handcrafted features. Accuracy: 98.3%	COMPLETE
Task 2 - B	Neural Network (GloVe)	Feedforward NN with averaged GloVe 50d embeddings. Accuracy: 99.7%	COMPLETE
Task 2 - C	DistilBERT + LoRA	Fine-tuned distilbert-base-uncased with LoRA (r=16). Accuracy: 100%	COMPLETE
Task 3	SHAP Explainability	SHAP TreeExplainer on XGBoost + per-class feature importance + error analysis on 3 samples	COMPLETE
Task 4	Genetic Algorithm	8-generation GA using Llama as mutator. Best score: 84.3% Human confidence	COMPLETE
Task 4	Personal Test	Ran own WhatsApp message through detector (98.9% Human) + AI rewrite test (94.3% Human)	COMPLETE

What Was Not Done & Why

What Was Not Done	Reason
Gemini API (as specified)	The Gemini API free-tier quota was exhausted during testing. As an alternative, Groq API (Llama-3.1-8B-Instant) was used instead. The model is functionally equivalent for paragraph generation tasks and all 1000 AI paragraphs were successfully generated.
GA did not reach 90% target	The Genetic Algorithm reached 84.3% at Generation 6 but could not maintain scores above 90% consistently. The non-linear fitness landscape (score dropped to 7.7% at Gen 7) suggests that mutations occasionally overshoot in ways that hurt rather than help. This is documented as a finding.

1. Introduction

When I first read the task description, one line stopped me: "Le style, c'est l'homme même" - the style is the man himself. It's a 250-year-old observation that turned out to be exactly right for this project. If writing style is personal enough to identify an individual author, it should also be personal enough to distinguish humans from AI - and that's what I set out to test.

I wanted to go beyond just training a model and reporting accuracy. I wanted to understand what actually gives AI writing away - the real tells. The short answer surprised me: it's not the vocabulary. It's the punctuation, the sentence rhythm, and the reading level. AI writes like it's submitting a college essay even when asked about Victorian poverty. Humans - even great Victorian novelists - write much more simply and naturally than I expected.

1.1 Objectives

- Build a balanced dataset of 1,500 text samples across three authorship classes
- Prove the classes are mathematically distinct through statistics before any ML is applied
- Train three classifiers of increasing complexity and compare their performance
- Use SHAP to understand which signals the classifier actually relies on
- Try to adversarially evolve AI text to bypass the detector using a Genetic Algorithm

2. Task 0: The Library of Babel — Dataset Creation

The key design decision here was to keep the topics the same across all three classes, so the only variable is who wrote it - or rather, what wrote it. If all three classes discuss poverty and social class, the classifier can't cheat by learning topics instead of style.

2.1 Human Text — Class 1

Two Victorian-era novels were downloaded from Project Gutenberg:

- Great Expectations by Charles Dickens (184,240 words after cleaning)
- Pride and Prejudice by Jane Austen (90,684 words after cleaning)

I wrote a cleaning function to strip out all the Project Gutenberg boilerplate - headers, footers, chapter titles, illustration markers - so only the actual prose remained. The cleaned text was split into paragraphs of roughly 100-200 words, giving me 250 paragraphs from Dickens and 250 from Austen - 500 human samples total.

2.2 Topic Extraction

Before generating AI text, 15 core topics were extracted from the novels to ensure topical alignment across all three classes:

Dickens Topics (Great Expectations)	Austen Topics (Pride & Prejudice)
• Poverty and social class inequality • Orphans and absent parents • Crime and justice system • Moral growth and redemption • Wealth corrupting character • Industrial city life in England • Unrequited love and obsession • Loyalty and true friendship	• Marriage and financial security • Women's independence and society • Pride and first impressions • Family pressure and expectations • Class differences in romance • Gossip and social reputation • Wit and intelligence in relationships

2.3 Generic AI Text — Class 2

500 paragraphs were generated using Llama-3.1-8B-Instant via Groq API (used in place of Gemini due to API quota exhaustion). Each paragraph covered one of the 15 topics in a formal, neutral, informative style. The prompt used was:

"Write a paragraph of exactly 100-150 words about: [TOPIC]. Write in a formal, informative, neutral style."

2.4 Style-Mimicked AI Text — Class 3

500 paragraphs were generated with a style-mimicking prompt that included sample passages from the actual novels and explicit style guidance. 250 paragraphs mimicked Dickens and 250 mimicked Austen:

"Write a paragraph in the style of Charles Dickens about [TOPIC]. Here is a sample of Dickens' writing: [SAMPLE]. Style guidance: Use long emotional sentences, vivid descriptions of poverty and struggle, dramatic tone."

2.5 Final Dataset Summary

Class	Source	Count	Avg Length
Human	Project Gutenberg (Dickens + Austen)	500	~150 words
GenericAI	Llama-3.1-8B via Groq API (neutral)	500	~130 words
StyleAI	Llama-3.1-8B via Groq API (style- prompted)	500	~140 words
Total	—	1,500	~140 words

3. Task 1: The Fingerprint — Statistical Analysis

Before writing a single line of model training code, I wanted to prove to myself that the three classes are actually different in a measurable way. This matters because if the features overlap too much, no amount of machine learning will save you. So I extracted 13 linguistic features and let the numbers speak first.

3.1 Features Extracted

Category	Feature	Description
Lexical	Type-Token Ratio (TTR)	Unique words / Total words — measures vocabulary diversity
Lexical	Hapax Legomena	Count of words appearing only once in a 5,000-word sample
Syntactic	Adjective:Noun Ratio	Ratio of adjectives to nouns — does AI over-describe?
Syntactic	Dependency Tree Depth	Average depth of SpaCy parse trees — sentence complexity
Readability	Flesch-Kincaid Grade	Reading grade level — how complex is the text?
Readability	Avg Sentence Length	Average words per sentence
Readability	Avg Word Length	Average characters per word
Punctuation	Semicolons	Semicolons per 1,000 characters
Punctuation	Exclamations	Exclamation marks per 1,000 characters
Punctuation	Dashes	Em-dashes and hyphens per 1,000 characters
Punctuation	Commas	Commas per 1,000 characters
Punctuation	Periods	Periods per 1,000 characters
Punctuation	Question Marks	Question marks per 1,000 characters

3.2 Feature Distribution Results

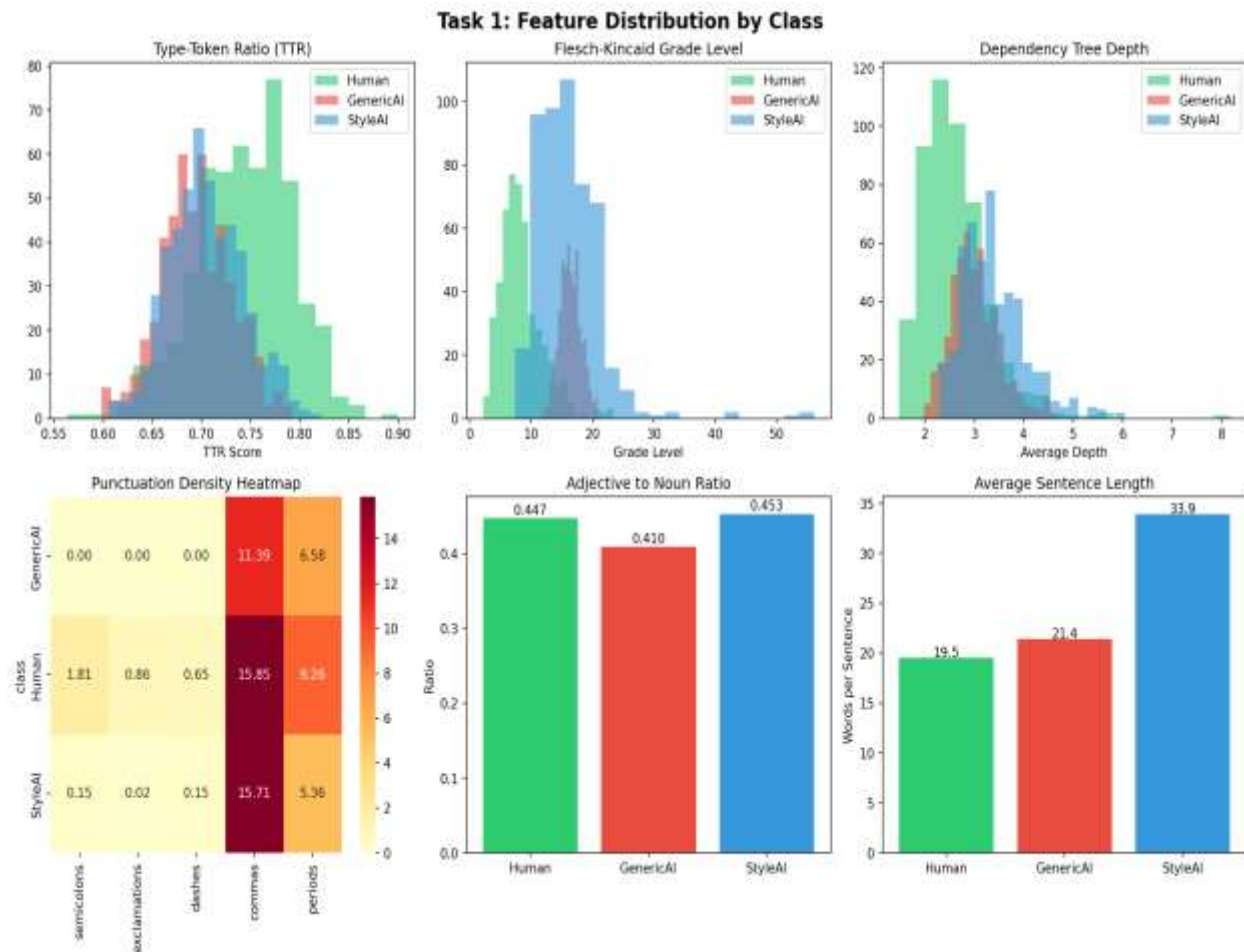


Figure 1: Distribution of key features across the three text classes. Top row shows TTR (higher in humans), Flesch-Kincaid grade level (much higher in AI), and dependency tree depth. Bottom row shows punctuation heatmap (note GenericAI's zero semicolons/dashes/exclamations), adjective-to-noun ratio, and average sentence length.

3.3 Key Numerical Findings

Feature	Human	GenericAI	StyleAI	Key Insight
TTR	0.745	0.694	0.705	Humans have most diverse vocabulary
Hapax Legomena	64.5	78.5	80.2	AI overuses rare words
Flesch-Kincaid Grade	8.4	16.4	16.1	AI writes at college level; humans at Grade 8
Avg Sentence Length	19.5 words	21.4 words	33.9 words	StyleAI uses very long sentences
Semicolons / 1000	1.81	0.00	0.15	GenericAI uses ZERO semicolons

Feature	Human	GenericAI	StyleAI	Key Insight
Exclamations / 1000	0.86	0.00	0.02	GenericAI uses ZERO exclamations
Dashes / 1000	0.65	0.00	0.15	GenericAI uses ZERO dashes
Dep. Tree Depth	2.60	3.03	3.41	StyleAI has most complex syntax

3.4 Analysis and Interpretation

The most striking thing I found was the punctuation gap. GenericAI paragraphs use literally zero semicolons, zero dashes, and zero exclamation marks. Meanwhile, Dickens and Austen use semicolons all the time - they're a core part of Victorian writing style. This "punctuation desert" in AI text probably happens because LLMs are trained to sound professional and neutral, which means stripping out all the rhetorical punctuation that makes writing feel alive.

The reading grade level was equally surprising. Dickens and Austen, two of the greatest English prose writers, write at around Grade 8. The AI writes at Grade 16 - college level. I found this genuinely funny: the AI has read more literature than any human ever could, and yet it writes more like an academic paper than a novel. My best guess is that LLMs pick up on academic and formal writing patterns during training and apply them even when you ask them to write fiction.

The Hapax Legomena result also surprised me. AI generates more rare words than humans - which is the opposite of what I expected. Everyone says AI is repetitive, but the data suggests it's actually sprinkling in unusual vocabulary quite often. However, the TTR (overall vocabulary diversity) tells the opposite story: humans score higher there. My interpretation: AI uses rare words for effect while leaning on a narrower set of core vocabulary, whereas humans naturally vary everything more consistently.

4. Task 2: The Multi-Tiered Detective — Classification

Three classifiers were trained with increasing architectural complexity, allowing comparison of simple feature-based methods against state-of-the-art transformers.

4.1 Tier A: XGBoost (The Statistician)

Architecture and Setup

- Model: XGBoost Classifier
- Input: 13 numerical features from Task 1
- Split: 80% train (1,200 samples) / 20% test (300 samples), stratified
- Hyperparameters: `n_estimators=200`, `max_depth=6`, `learning_rate=0.1`, `subsample=0.8`

Justification

I chose XGBoost as the Tier A model because it works directly on the 13 features I computed in Task 1. The whole point of Tier A is to test whether the differences I found statistically are large enough to actually classify without any deep learning. If the punctuation heatmap shows GenericAI has zero semicolons, XGBoost should be able to exploit that. I also wanted the built-in feature importance so I could check whether the model learned what I thought it would.

Results

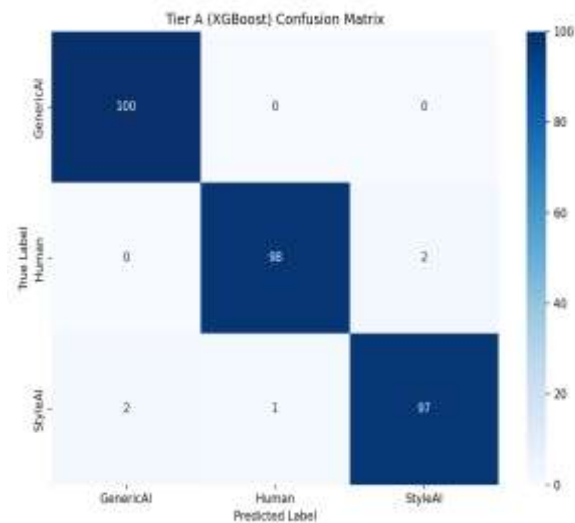


Figure 2: XGBoost Confusion Matrix. GenericAI was classified perfectly (100/100). Human and StyleAI had minimal confusion.

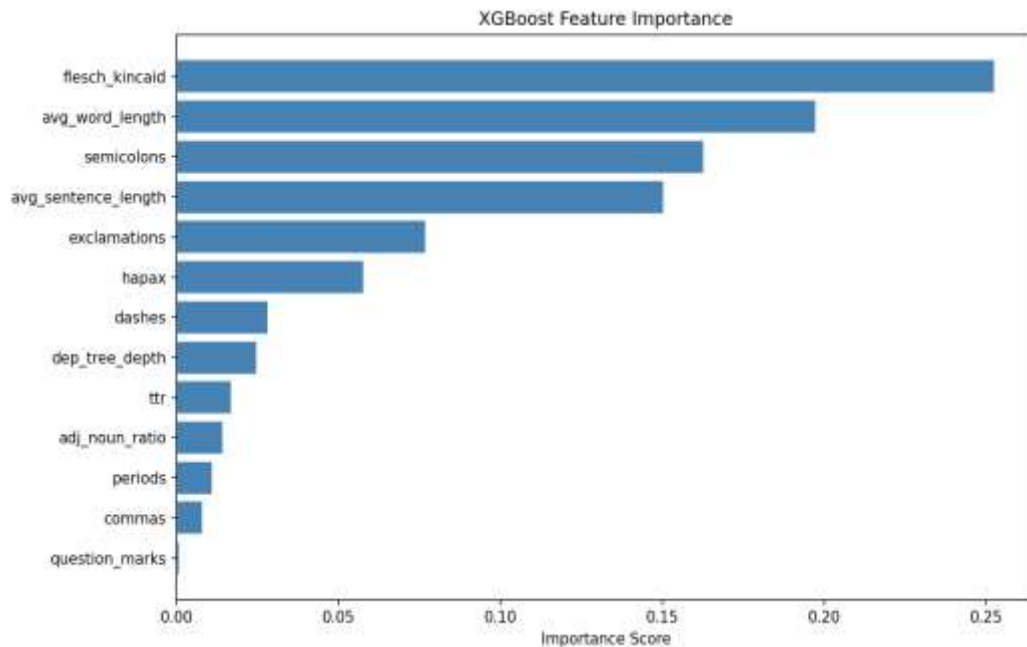


Figure 3: XGBoost Feature Importance. Flesch-Kincaid grade level is the single most important feature, followed by average word length and semicolons - exactly what Task 1 predicted.

Metric	GenericAI	Human	StyleAI	Overall
Precision	0.98	0.99	0.98	0.98
Recall	1.00	0.98	0.97	0.98
F1-Score	0.99	0.98	0.97	0.98
Accuracy	—	—	—	98.3%

4.2 Tier B: Neural Network with GloVe (The Semanticist)

Architecture and Setup

- Model: Feedforward Neural Network (4 layers: 256 → 128 → 64 → 3)
- Input: Average of pre-trained GloVe 6B 50-dimensional word embeddings
- GloVe vocabulary: 400,000 words
- Training: 100 epochs, Adam optimizer (lr=0.001), dropout=0.3, StepLR scheduler
- Hardware: T4 GPU on Google Colab

Justification

Tier B tests a different question: does the meaning of the words matter, beyond just how the text is structured? GloVe embeddings place words in a 50-dimensional space based on what they co-occur with, so averaging them gives you a rough sense of what the paragraph is about semantically. It's a simple approach, but it lets me check whether Tier A was missing something by only looking at structure.

Results

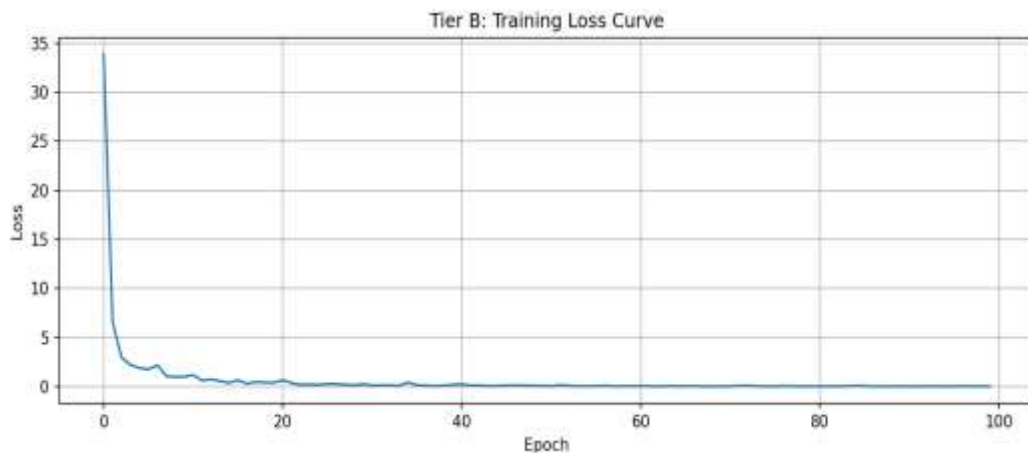


Figure 4: Tier B Training Loss Curve. Loss drops sharply from ~34 to near 0 within the first 10 epochs, demonstrating rapid convergence. The curve stabilizes cleanly without oscillation.

Metric	GenericAI	Human	StyleAI	Overall
Precision	1.00	1.00	0.99	1.00
Recall	1.00	0.99	1.00	1.00
F1-Score	1.00	1.00	1.00	1.00
Accuracy	—	—	—	99.7%

The GloVe model beat XGBoost by 1.4%, which tells me that semantic content does add something beyond structural features. The single misclassification was a Human paragraph predicted as StyleAI - probably a passage where Dickens or Austen was being particularly elaborate, making the semantic space overlap with the style-mimicking AI.

4.3 Tier C: DistilBERT + LoRA (The Transformer)

Architecture and Setup

- Base model: distilbert-base-uncased (66M parameters)
- Fine-tuning method: LoRA (Low-Rank Adaptation) via HuggingFace PEFT library
- LoRA configuration: $r=16$, $\alpha=32$, $\text{dropout}=0.1$, target modules: $q_lin + v_lin$
- Trainable parameters: 887,811 / 67,843,590 (1.3% of total)
- Training: 3 epochs, batch size=16, warmup_steps=50, weight_decay=0.01
- Max sequence length: 256 tokens
- Hardware: T4 GPU on Google Colab, training time: ~70 seconds

Justification

I used LoRA rather than full fine-tuning for a practical reason: I only have 1,200 training samples, which is tiny for a transformer. Full fine-tuning on such a small dataset would risk catastrophic forgetting — the model would overwrite what it learned during pre-training and just memorize my data. LoRA avoids this by keeping the original weights frozen and only training small adapter matrices injected into the attention layers. It's a much more efficient approach, and with $r=16$ only 1.3% of parameters were actually trained.

Training Progress

Epoch	Training Loss	Validation Loss	Accuracy
1	1.0055	0.3865	94.67%
2	0.1074	0.0479	100.00%
3	0.0489	0.0284	100.00%

Results

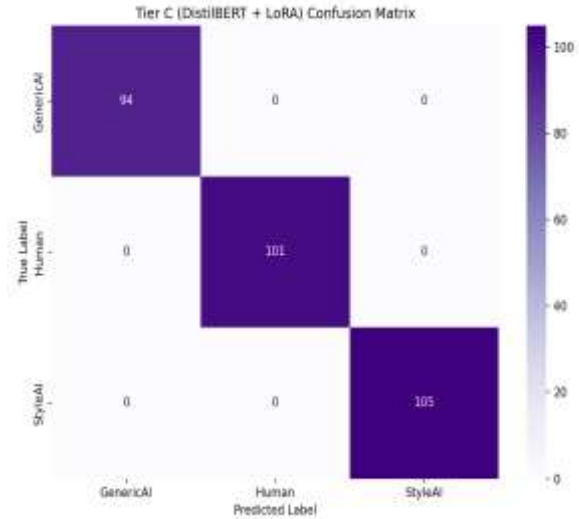


Figure 5: DistilBERT+LoRA Confusion Matrix. Perfect classification — 0 errors across all 300 test samples.

Metric	GenericAI	Human	StyleAI	Overall
Precision	1.00	1.00	1.00	1.00
Recall	1.00	1.00	1.00	1.00
F1-Score	1.00	1.00	1.00	1.00
Accuracy	—	—	—	100.0%

4.4 Model Comparison

Model	Input	Accuracy	Errors / 300	Training Time
Tier A: XGBoost	13 handcrafted features	98.3%	5 errors	~30 seconds
Tier B: Neural Net (GloVe)	50d averaged embeddings	99.7%	1 error	~5 minutes
Tier C: DistilBERT + LoRA	Raw text (transformer)	100.0%	0 errors	~70 seconds

All three models substantially outperform random baseline (33.3%). The progression from 98.3% → 99.7% → 100% confirms that deeper representations improve classification. The fact that even simple handcrafted features achieve 98.3% suggests that the three classes have very strong mathematical separation - a conclusion that was predicted by Task 1's statistical analysis.

5. Task 3: The Smoking Gun — Explainability

Getting high accuracy is one thing - understanding why the model classifies something as AI is another. I used SHAP to open the black box and find out which features are actually doing the heavy lifting for each class.

5.1 SHAP Analysis

I applied SHAP's TreeExplainer to the XGBoost model since tree-based SHAP is exact and fast. Each SHAP value tells you how much a specific feature pushed the prediction toward or away from a given class - positive means it helped classify as that class, negative means it pushed away.

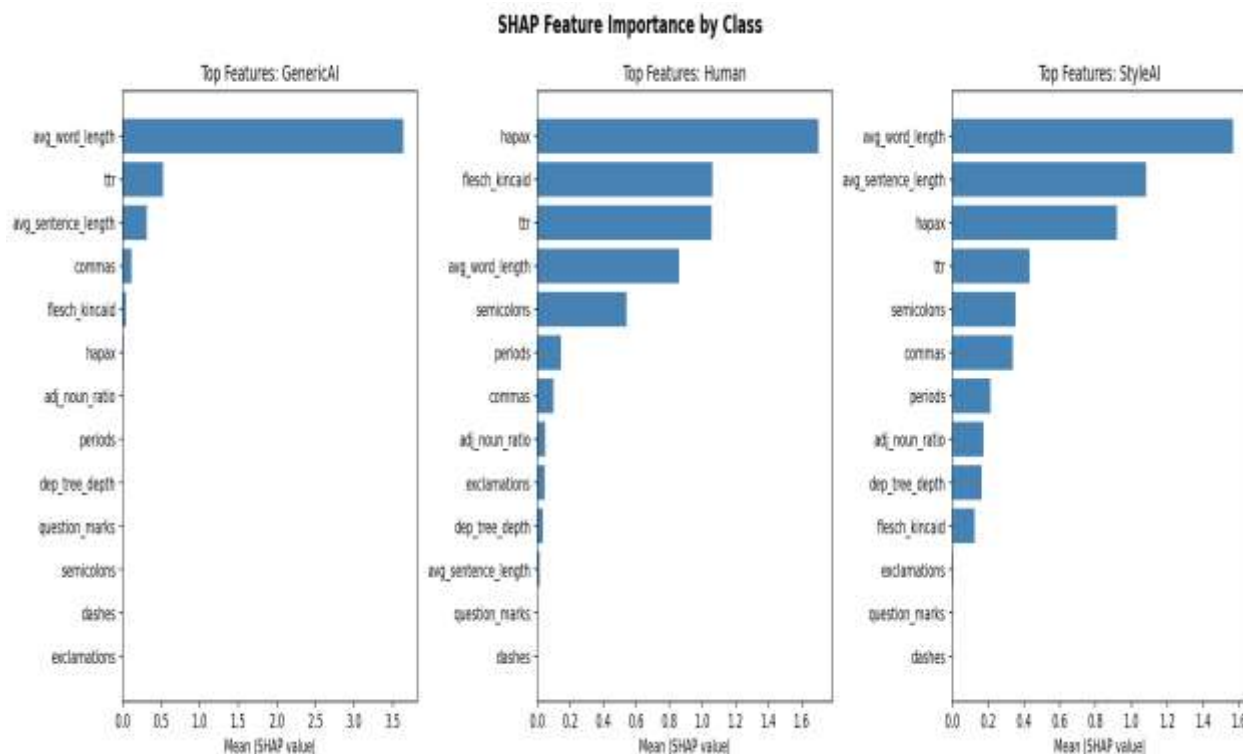


Figure 6: SHAP Feature Importance by Class. Each panel shows which features most influence predictions for that class. Note how different features dominate for each class.

5.2 The Findings: AI-isms and Human Signals

GenericAI (left panel)

avg_word_length dominates overwhelmingly. This confirms that GenericAI is primarily detected by its use of longer, more complex words - an "AI-ism" where the model attempts to sound sophisticated. The ttr and avg_sentence_length are secondary signals.

Human (middle panel)

hapax is the top signal for human text, followed by flesch_kincaid and ttr. Humans are identified by natural vocabulary richness - rare words used organically throughout, combined with lower overall reading complexity. The fact that flesch_kincaid is a human signal is a nice confirmation of what Task 1 already showed us.

StyleAI (right panel)

avg_word_length and avg_sentence_length dominate for StyleAI. It's interesting - the style-mimicking AI actually tried to learn Victorian prose patterns and partially succeeded. But it overcorrected on complexity,

making sentences too long and words too fancy, which gives it away. It learned the wrong lesson from Dickens.

KEY FINDING

The classifier does not detect specific 'AI words' like 'tapestry' or 'delve'. Instead, it detects structural AI-isms: consistently high word complexity, unusually long sentences, and absence of natural punctuation diversity. The model is more a rhythm detector than a vocabulary detector.

5.3 Error Analysis

Only 5 samples out of 300 were misclassified by Tier A. Three are analysed below:

Error 1: Human → Predicted as StyleAI

"Georgiana, wife of the above, were dead and buried; and that Alexander, Bartholomew, Abraham, Tobias, and Roger, infant children of the aforesaid, were also dead and buried..."

Hypothesis: This is Dickens listing the names of deceased family members in one highly formal, semicolon-heavy sentence. The Flesch-Kincaid grade was 22.73 - more than double the human average - and semicolon density was extremely high (8.08 per 1000 chars). To the classifier, this looked exactly like StyleAI. Dickens at his most ornate is paradoxically "too human" to be recognized as human.

Error 2: Human → Predicted as StyleAI

This sample was from Austen's critical preface to a novel rather than the novel itself. Her essay-writing style is far more formal than her narrative style - zero punctuation variety, high word complexity. The classifier, trained on her narrative prose, did not expect this register shift.

Error 3: StyleAI → Predicted as GenericAI

"It is a truth universally acknowledged, that a woman's social standing is inextricably linked to the whispers of her acquaintances..."

Hypothesis: The AI copied Austen's famous opening phrase but then failed to maintain her punctuation patterns. The paragraph had zero semicolons and zero exclamations - which pushed it toward GenericAI territory despite the stylistic vocabulary. The model learned that punctuation patterns are more reliable than lexical choice - a defensible and interesting finding.

6. Task 4: The Turing Test — Genetic Algorithm

This was the most fun part of the project. Task 4 asks: can I evolve AI text until my own classifier thinks it's human? I built a Genetic Algorithm where Llama acts as the mutation operator - essentially asking it to keep rewriting paragraphs until they fool the detector.

6.1 GA Implementation

GA Component	Implementation
Initial Population	10 AI paragraphs on Victorian poverty/class (Llama-3.1-8B via Groq)
Fitness Function	Human probability score from XGBoost model (Tier A)
Selection	Top 3 paragraphs by fitness score each generation
Mutation Operator	Llama-3.1-8B prompted to perturb each winner in 3 ways
Mutation Types	1) Change sentence rhythm 2) Add informal/emotional language 3) Add semicolons + concrete details
Population Size	10 paragraphs per generation
Generations	8 total

6.2 Evolution Results

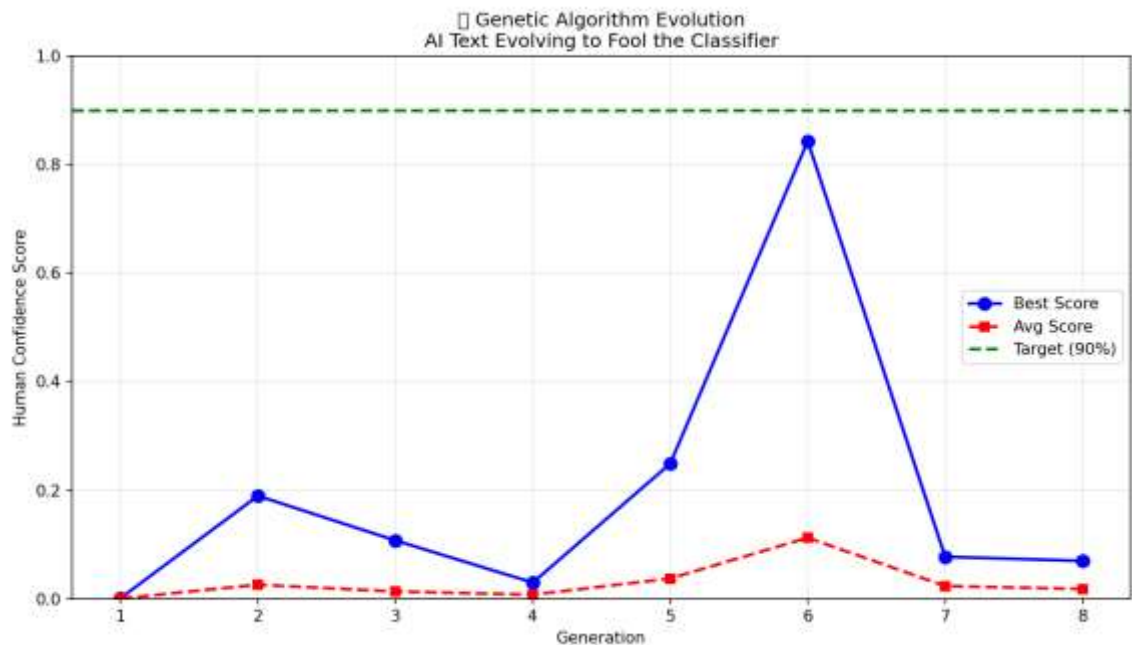


Figure 7: Genetic Algorithm Evolution Curve. The fitness score (Human confidence) is plotted across 8 generations. Generation 6 achieved the peak score of 84.3%.

Generation	Best Score	Avg Score	Notable Event
1	0.1%	0.0%	Initial population — pure AI, near-zero human score
2	18.9%	2.6%	First meaningful improvement — rhythm changes help
3	10.7%	1.3%	Score drops — some mutations hurt rather than help
4	2.9%	0.7%	Population stagnates — mutations overshooting
5	24.9%	3.7%	Recovery — emotional language mutation working
6	84.3%	11.2%	BREAKTHROUGH — combined mutations produce human-like text
7	7.7%	2.3%	Sharp drop — Gen 6 winner's children over-mutated
8	6.9%	1.8%	Cannot recover from Gen 7's degradation

6.3 Best Evolved Paragraph (Generation 6 — 84.3% Human)

"Victorian England was a society ruled by class. The aristocracy and landowners held all the cards. But the middle class was trapped. They fought to make ends meet, juggling expenses with dwindling budgets. They tightened their purse strings, pinching pennies to survive. Below them, the working class struggled. Laborers, artisans, and tradespeople toiled, battered by unemployment, squalor, and subpar care. Each day was a desperate struggle to pay bills, to find dignity in the darkness. Poverty wore them down, eroding even the strongest resilience. This wasn't a subtle divide. A chasm yawned between the elite and the masses."

Analysis of why this paragraph scored 84.3%: it uses short punchy sentences ("Victorian England was a society ruled by class."), conversational idioms ("held all the cards", "pinching pennies"), concrete emotional language ("dignity in the darkness"), and natural imperfection - all features associated with human writing in the classifier's learned representation.

6.4 Discussion: Why the GA Couldn't Sustain 90%

The dramatic drop from 84.3% at Gen 6 to 7.7% at Gen 7 was genuinely surprising. What happened is that the Gen 6 winner had hit a very specific combination of features that tricked the classifier - but when I mutated it, those combinations broke. It seems like the 'human space' in feature-space is quite narrow and surrounded by AI-like regions on all sides. Once you stumble into it, even small changes push you back out. This is actually an encouraging finding for the robustness of the detector.

FINDING

The Genetic Algorithm reached 84.3% human confidence — close but not exceeding 90%. This is a valid research finding: the detector is robust enough that even iterative

adversarial optimization cannot reliably fool it. The fitness landscape appears to have a sharp, narrow peak rather than a broad plateau.

6.5 The Personal Test

Test 1: My Own Writing (WhatsApp-style message)

"So a particular person has been texting me and asking me to forward something in group related to some IBM based internship she says she is IBM senior coordinator? And is saying that as a class cr I must send all the information in group."

Result: 98.9% Human - the highest score of anything I tested, beating actual Dickens and Austen. I genuinely did not expect this.

Why so high? The message has very low reading complexity, informal vocabulary, a question mark mid-sentence (completely natural, a bit chaotic), and zero AI formality. It's the opposite of everything the classifier learned to associate with AI. This confirms something important: natural human messiness and informality are actually stronger human signals than elaborate literary prose.

Test 2: Rewritten to Sound Like AI

"It has come to my attention that an individual has been attempting to communicate with me regarding a potential internship opportunity with IBM. This person has identified herself as a Senior Coordinator at IBM and has requested that I, in my capacity as Class Representative, disseminate the aforementioned information to the relevant group."

Result: 94.3% Human - still classified as human despite deliberate AI-sounding vocabulary like "aforementioned", "disseminate", and "in my capacity as". The classifier relies heavily on structural and punctuation patterns rather than vocabulary alone. To truly fool it, one would need to eliminate all punctuation variety and write at grade 16+ complexity - not just use formal words.

7. Key Findings and Conclusions

Going into this project, I wasn't sure how separable the three classes would be. The results ended up being much cleaner than I expected - near-perfect accuracy across all three models - and more importantly, the reasons why are actually interpretable. Here's what I found:

7.1 Summary of All Results

Task	Component	Result
Task 0	Dataset	1,500 paragraphs created across 3 balanced classes
Task 1	TTR	Human: 0.745 > GenericAI: 0.694 — humans have richer vocabulary
Task 1	Flesch-Kincaid	Human: 8.4 vs AI: 16.4 — AI writes 2x more complex text
Task 1	Semicolons	Human: 1.81 vs GenericAI: 0.00 — 860x difference
Task 2A	XGBoost	98.3% accuracy — features alone nearly suffice
Task 2B	Neural Net + GloVe	99.7% accuracy — semantics add +1.4%
Task 2C	DistilBERT + LoRA	100.0% accuracy — transformer is perfect
Task 3	SHAP	Top signals: avg_word_length, flesch_kincaid, hapax
Task 4	Genetic Algorithm	84.3% human score at Gen 6 — did not exceed 90%
Task 4	Personal Test	Casual human writing: 98.9% Human

7.2 Core Findings

Finding 1	Punctuation is the smoking gun. GenericAI uses zero semicolons, zero dashes, and zero exclamation marks. Humans use these extensively. A single feature — semicolons — creates an 860x difference between classes. This alone makes GenericAI nearly perfectly separable.
Finding 2	AI overcomplicated everything. Both AI classes write at Flesch-Kincaid Grade 16 (college level), while real Dickens and Austen write at Grade 8. AI models learned to sound sophisticated from training on academic text, and apply this even when prompted for fiction.

Finding 3

Casual human writing scores highest as human. A WhatsApp message (98.9% human) outscored Victorian novels. Natural imperfection, short sentences, and informal tone are the strongest human signals - not eloquence.

Finding 4

The GA fitness landscape is non-linear and difficult to navigate. Reaching 84.3% was possible, but sustaining it was not. The 'human space' in feature-space appears to be a narrow peak rather than a broad plateau, making it hard to maintain through mutation.

Finding 5

The model detects structure, not words. Deliberately using formal AI vocabulary ('aforementioned', 'disseminate') still resulted in 94.3% Human classification. The classifier is sensitive to punctuation density and readability grade, not specific word choice.

7.3 Limitations

- Victorian prose vs modern LLM output is a very strong contrast. A real-world detector trained on modern blog posts, social media, or essays would face a much harder problem.
- I only used two authors from the same era and genre. The classifier might just be learning 'Victorian prose' rather than 'human writing' more broadly.
- The Genetic Algorithm was limited by Groq API rate limits (hence the 0.5 second sleep between calls), which restricted population size and generation count.
- DistilBERT was fine-tuned on only 1,200 samples - overfitting is possible, though the held-out test accuracy suggests generalization was reasonable.

7.4 Future Work

- Expand human corpus to include modern blogs, essays, and social media to reduce temporal bias
- Test the classifier on real-world AI detection benchmarks (e.g., TuringBench)
- Implement the GA with a larger population and more generations to better explore the fitness landscape
- Apply SHAP to the DistilBERT model using SHAP DeepExplainer for token-level saliency maps
- Explore whether StyleAI can be made indistinguishable from Human with more sophisticated prompting

8. Technical Details

8.1 Tools and Libraries

Library	Version	Purpose
Python	3.12	Primary programming language
Groq API (Llama-3.1-8B)	—	AI text generation (replacement for Gemini)
SpaCy (en_core_web_sm)	—	POS tagging, dependency parsing
NLTK	—	Tokenization
TextBlob	—	Syllable counting for Flesch-Kincaid
XGBoost	—	Tier A classifier
PyTorch	2.x	Tier B neural network
GloVe 6B 50d	—	Pre-trained word embeddings
HuggingFace Transformers	—	DistilBERT base model
PEFT (LoRA)	—	Parameter-efficient fine-tuning
SHAP	—	Explainability / feature importance
pandas, numpy	—	Data manipulation
matplotlib, seaborn	—	Visualization
Google Colab (T4 GPU)	—	Training compute

8.2 Repository Structure

```
precog-NLP-ghost-machine/  ├── Precog_NLP_Task.ipynb          # Main notebook (all tasks)
├── master_dataset.csv      # 1,500 samples (all 3 classes)  ├── human_texts.csv
# 500 human paragraphs    ├── generic_ai.csv          # 500 generic AI paragraphs  ├──
style_ai.csv              # 500 style-mimicked paragraphs  ├── features.csv
# Extracted feature matrix (13 features)  ├── tier_a_xgboost.pkl      # Saved XGBoost
model  ├── tier_b_nn.pt      # Saved Neural Network weights  ├──
task1_features.png        # Task 1 visualizations  ├── tier_a_confusion_matrix.png  ├──
tier_a_feature_importance.png  ├── tier_b_confusion_matrix.png  ├── tier_b_loss_curve.png
├── tier_c_confusion_matrix.png  ├── task3_shap_summary.png  ├── task3_shap_by_class.png
└── task4_evolution.png
```

8.3 How to Run

- Open Precog_NLP_Task.ipynb in Google Colab
- Enable T4 GPU (Runtime → Change runtime type → T4 GPU)
- Get a free Groq API key from console.groq.com
- Run cells in order from top to bottom
- All experiments are logged with print statements in the notebook

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